Section 02

Industrial Security (Master)

IS Security

Section 02 – PLC Ladder Logic Reverse Engineering

PLC Ladder Logic Reverse Engineering

Lecture Notes

Department of Computer Science

Outline – Section 02

1. Why Ladder RE
2. Acquisition
3. Decoding
4. What to Look For
5. Defending the Code

Learning Objectives

🎓 By the end of this section you will be able to

- Extract a ladder-logic program from a PLC and from a project file.
- Recognise vendor-specific block formats (Siemens, Rockwell, CODESYS).
- Decompile or visualise ladder logic and identify what the program does.
- Spot logic-bomb and stealth modifications (à la Stuxnet).
- Apply integrity controls to prevent unnoticed code change.

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Why Ladder RE

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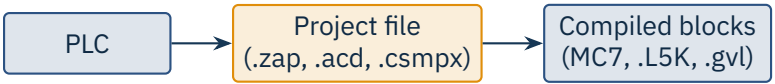
Why Ladder Logic Reverse Engineering?



The blind spot

The PLC runs the *compiled* blocks. The vendor tool shows the *project*. If an attacker changes only the runtime, the project still looks fine and only RE catches it.

Source: Stuxnet (2010) is the canonical example – HMI fed clean data while the PLC ran tampered logic.

Three Common Vendor Worlds

	Siemens S7-3xx/4xx	Rockwell ControlLogix	CODESYS family
Project file	`.zap`, `.s7p` (Step7); `.ap16` (TIA)	`.acd`	`.project` (XML zip)
On-PLC format	MC7 byte-code, OBxx/FCxx/DBxx blocks	`.L5K` compiled tags	UDFB structured text
Read-back tool	`snap7`, TIA upload	RSLogix5000 upload, `libplctag`	CODESYS device tree

Source: Snap7 docs; *libplctag*; Rockwell PCDC; CODESYS GmbH.

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Acquisition

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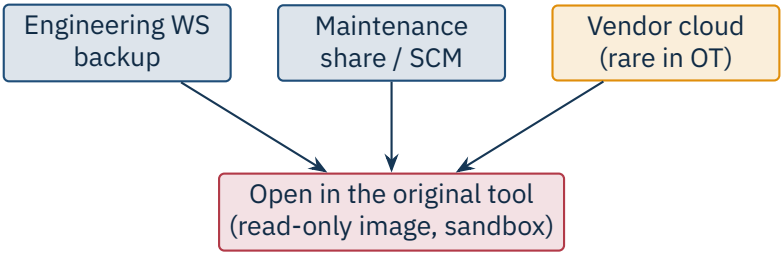
Reading Blocks via Snap7 (Siemens)


```
import snap7
c = snap7.client.Client()
c.connect("10.0.0.10", 0, 1)
# List blocks, then upload OB1 + DB1
blocks = c.list_blocks()                # SfcCount, OBCount, FCCount...
ob1 = c.upload(snap7.types.Block.OB, 1) # raw MC7 bytes
db1 = c.upload(snap7.types.Block.DB, 1)
open("ob1.bin", "wb").write(ob1)
```

Permission check

The S7-300/400 default *put/get* setting allows upload; modern S7-1500 requires explicit enable. Document any change.

Project-File Acquisition



Always sandbox

Project files have run macros and *network ranges*. Open them in a disconnected VM with no host shares.

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Decoding

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Siemens MC7 Block Layout

Tools

`s7-pcap` and `step7-parser` (Sergey Gordeychik et al., 2015) parse MC7. Newer S7commPlus blocks (S7-1200/1500) use `Pdata` and require S7commPlus-aware tools.

Source: Sergey Gordeychik et al., *SCADA StrangeLove*, 2015 | Beresford, D., *Exploiting Siemens Simatic S7 PLCs*, 2011.

Rockwell .L5K and .ACD

- `.acd` is a binary container; `.l5k` / `.l5x` is the printable export.
- `l5k_parser` (community) reconstructs Structured Text from the export.
- Watch for **add-on instructions (AOIs)** – compiled inside the project; their internals are hidden in the GUI.
- Tag aliases obscure semantics: `Bool15` may secretly be “open_valve”.

Decompile then re-render

Round-trip the export through RSLogix; what visually matches the original is faithful, what diverges is your target.

CODESYS / IEC 61131-3 Round-Trip

CODE16 (2023)

Sixteen CVEs in CODESYS Runtime exposed authentication, parser, and update-verifier flaws across hundreds of vendor products built on the runtime.

Source: CISA *ICSA-23-103-09* and related advisories; Microsoft *CoDe16* disclosure, 2023.

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What to Look For

The Stuxnet Pattern

1. Pristine OB1 / main loop sets up normal control

2. Hidden OB35 (cyclic) checks for the target signature (frequency, set-point)

3. On match: record current set-points, replace with malicious sequence

4. When operator reads back, recorded set-points are returned (HMI lied)

Source: Langner, R., *To Kill a Centrifuge*, The Langner Group, 2013.

Logic Bomb Indicators

- Unused inputs that are read but never appear in the rung logic.
- Time- or counter-based branches that never fire under normal data.
- Blocks named after vendor utilities (“OEM_diag”, “factory_test”) outside the official block list.
- Constants whose values match *exactly* known process set-points.
- Calls into reserved or vendor-locked memory areas.
- Discrepancy between project-file timestamps and on-PLC block CRCs.

Caution

A missing CRC, or a CRC that does not match the project file, is more often vendor sloppiness than malice – but it deserves a written investigation.

Tagging the Memory Map

Reverse engineering tip

Cross-reference *every* symbol against where it is read and where it is written. Symbols that are only *written* (never read) are smoke.

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Defending the Code

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Defensive Controls

- Signed firmware + signed program downloads (vendor support varies)
- PLC key-switch in “RUN” position; “PROG” requires physical access
- Continuous block-checksum monitoring (Claroty, Dragos, Nozomi modules)
- Version-controlled project files with CI & review
- Engineering workstation hardened; no Internet, MFA, application allow-listing

Disclosure Considerations

- Findings on production logic touch **safety** – coordinate with process safety, not just IT.
- Vendor PSIRT first; CERT/CC second; public publication only after fix.
- Sharing tags or constants from real plants risks identifying the asset owner – redact heavily.
- Some jurisdictions (US DMCA §1201, EU CRA research safe-harbour, German §202c StGB) constrain RE; consult counsel.

Caution

Logic from a production PLC you do not own is contaminated evidence. No serious vendor will accept it.

Summary – Section 02

★ Key Takeaway: Ladder RE for ICS

- PLCs run compiled blocks, not the project source – they can diverge silently.
- Project-file and runtime acquisition each catch different attacks.
- Vendor-specific tools (snap7, l5k-parser, CODESYS) are the workhorses.
- Stuxnet-style logic bombs hide in unused inputs, time bombs, and CRC mismatches.
- Defence is signed downloads, key-switch, continuous checksum monitoring.

References – Section 02

- Langner, R. *To Kill a Centrifuge*, The Langner Group, Nov 2013.
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- Snap7 project. `https://snap7.sourceforge.net`.
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- Dragos, Claroty, Nozomi. PLC integrity-monitoring product documentation.

End of Section 02

PLC Ladder Logic Reverse Engineering

Questions and discussion

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